

HEAP

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Heap is a complete binary tree. There are two types of heapes as max heap and min heap.

Max-Heap: If the value present at any node is greater than all its children then such a tree is called as the max-heap.

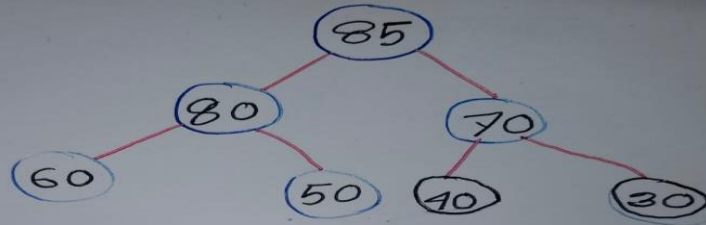


Fig: A Max heap

Min-Heap: If the value present at any node is smaller than all its children then such a tree is called as the min-heap.



Fig: A min heap

Heap Creation

We add elements one by one and place them in respective positions.

1. Add an element as first element(root) of the heap.
2. Select a child and place it in proper position.
3. Add next element in left of the root and compare with root element.
4. Add next element in right of the root and compare with root element and place the right position.
5. Get a Heap as an output

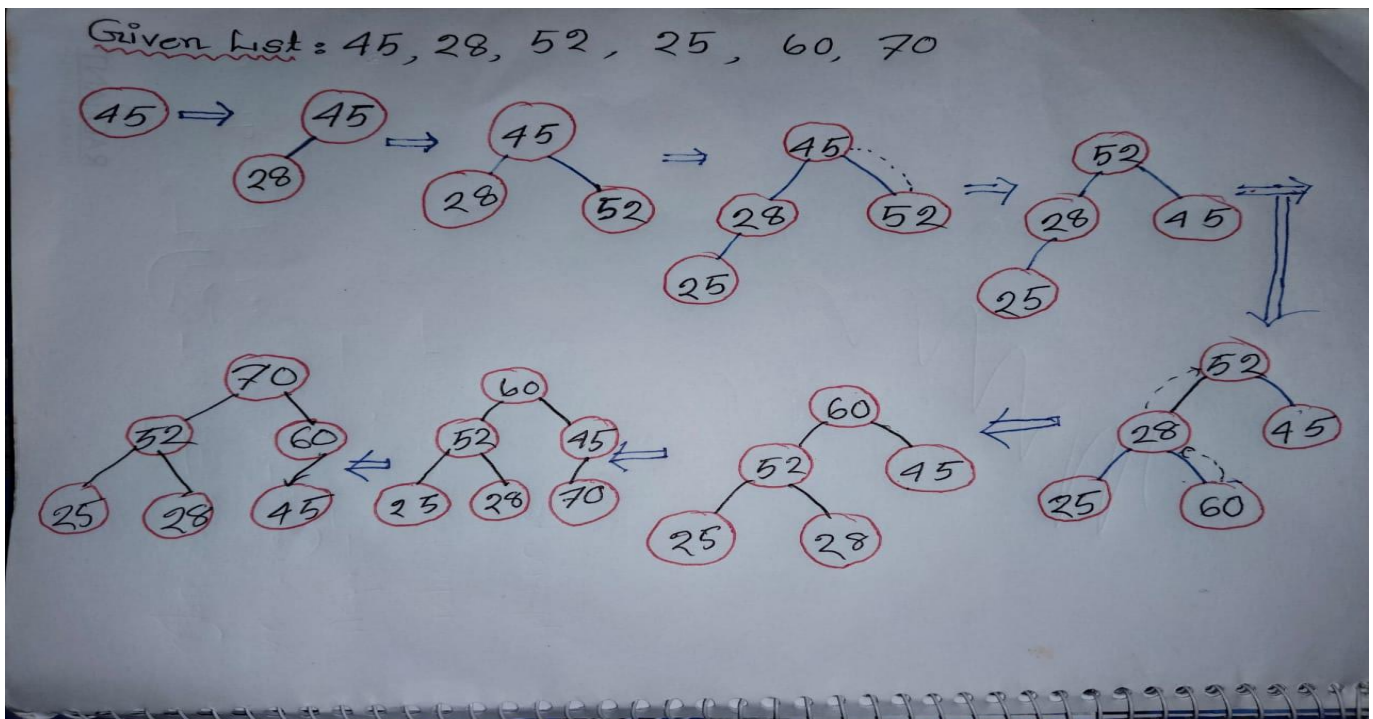


Fig: Heap